Auteur: Abdellah El Moussaoui
Studierichting: Informatica
Oefeningenreeks 4A nr. 42.19

$$\int \frac{2x^4-5x^3+x^2+6x-7}{x^2-3x+2}dx$$
$$\Rightarrow \frac{2x^4-5x^3+x^2+6x-7}{x^2-3x+2} = 2x^2+x+\frac{4x-7}{x^2-3x+2}$$
$$= 2x^2+x+\frac{4x-7}{(x-1)(x-2)}$$
$$\frac{4x-7}{(x-1)(x-2)} = \frac{A}{x-1}+\frac{B}{x-2}$$
$$A = \left.\frac{4x-7}{x-2}\right|_{x=1} = \frac{-3}{-1} = 3$$
$$B = \left.\frac{4x-7}{x-1}\right|_{x=2} = 1$$
$$\Rightarrow \int\left(\frac{3}{x-1}+\frac{1}{x-2}\right)dx$$
$$= 3\ln|x-1|+\ln|x-2|+k$$
$$\boxed{\int \frac{2x^4-5x^3+x^2+6x-7}{x^2-3x+2}dx = \frac{2}{3}x^3+\frac{1}{2}x^2+3\ln|x-1|+\ln|x-2|+k}$$

References